

A Linearized CGE Approach to a Real-Time Sectoral–Regional Layoff-Risk Dashboard for Indonesia: Methodology and Data Retrieval Pipeline

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Contents

1	Purpose and Overview	2
1.1	The Core Design Problem	2
1.2	Three Shock Themes	2
2	Notation	3
3	Sector and Region Classification	3
3.1	The 52-Sector Aggregation	3
3.2	Provinces	4
4	Theme 1: Commodity Price Shocks	4
4.1	Mechanism	4
4.2	Commodity List and Data Sources	4
4.3	Shock Magnitude Construction	5
4.4	Elasticity Calibration	5
5	Theme 2: Trading-Partner Growth Shocks	6
5.1	Mechanism	6
5.2	Step 1: Country Growth Revisions	6
5.3	Step 2: Export-Share Weights	7
5.4	Step 3: Aggregation into a Sector-Specific Shock	7
5.5	The Tradable Sector Set $\mathcal{K}$	7
5.6	Elasticity Calibration	8
6	Theme 3: Domestic Demand Shocks	8
6.1	Mechanism and Motivation	8
6.2	Why a Real (Volume) Index, Not a Price Index	8
6.3	Data Source	9
6.4	“Press Update” Logic for Theme 3	9
6.5	IPR Categories	9
6.6	Why Year-on-Year, Not Month-to-Month	10
6.7	Benchmark/Trend and Shock Construction	10

6.8	Regional Dimension: National Shock, Regional Elasticity	10
6.9	Mapping IPR Categories to the 52-Sector Classification	11
6.10	Elasticity Calibration: All 52 Sectors	11
7	Why Elasticities Are Theme-Specific	12
8	Aggregation Across Themes	12
9	Severity-Weighted Heatmap	13
10	Computational Architecture: “Press Update” Logic	13
11	Assumptions and Known Limitations	14
12	Future Work	15
13	Open Items Requiring Further User Input	15

1 Purpose and Overview

This document specifies the methodology and complete data-retrieval pipeline for a dashboard that maps current economic conditions into a **52-sector** $\times$ **34-province** matrix of employment change, used as an early-warning layoff-risk signal for Indonesia. The dashboard is intended for use by Indonesia’s Ministry of Manpower (Kemnaker) and is built on IndoTERM, a computable general equilibrium (CGE) model of the Indonesian economy.

1.1 The Core Design Problem

A full CGE solve is computationally expensive and cannot be run live every time a user presses “update.” The design solves this by separating two steps with very different cost profiles:

1. **Offline calibration** (expensive, infrequent): IndoTERM is solved once per shock, at a small (1%) perturbation, to obtain a calibrated elasticity — the percentage employment response in every sector–province cell to a unit shock. This step is run by the user outside the dashboard and only needs to be repeated when the model, closure rules, or base-year Social Accounting Matrix (SAM) change.
2. **Live aggregation** (cheap, every refresh): on each dashboard update, the system retrieves the latest values of various real-world economic indicators, converts them into shock magnitudes, and multiplies them against the pre-calibrated elasticities — a simple tensor contraction, not a CGE solve.

This is a deliberate, documented shortcut, not an oversight. Section 11 states explicitly which assumptions this rests on, and Section 12 describes the planned transition away from this shortcut once server-side, on-demand CGE solving becomes feasible.

1.2 Three Shock Themes

Three conceptually distinct families of shocks feed into the same sector–province matrix:

Theme 1 — Commodity Price Shocks. World/domestic price changes in eight key Indonesian export and import commodities, directly affecting the sectors that produce or use them.

Theme 2 — Trading-Partner Growth Shocks. Revisions to IMF growth forecasts for Indonesia’s trading partners, translated into export-demand shocks for tradable sectors via each sector’s actual country-by-country export exposure.

Theme 3 — Domestic Demand Shocks. Deviations of domestic retail sales volume from trend, by category, capturing slowdowns in domestically-oriented sectors that the first two themes, being externally focused, do not.

Each theme constructs its own shock magnitudes from different real-world data sources (detailed in Sections 4–6) and is multiplied against its own, separately-calibrated elasticity tensor (a deliberate modeling choice explained in Section 7). The three themes’ contributions are then summed (Section 8) into a single severity-weighted heatmap (Section 9).

2 Notation

Symbol	Meaning
i	Sector index, $i = 1, \dots, 52$ (IndoTERM’s AGGIND aggregation)
r	Province index, $r = 1, \dots, 34$
j	Generic shock index within a theme
θ	Theme index, $\theta \in \{1, 2, 3\}$
$\eta_{i,r,j}^{(\theta)}$	Calibrated elasticity: % change in employment in cell (i, r) induced by a 1% shock to driver j under theme θ , holding all other shocks at zero
x_j	Live shock magnitude for driver j , the dashboard’s run-time input
$E_{i,r}^{(\theta)}$	Theme θ ’s contribution to total % employment change in cell (i, r)
$E_{i,r}$	Total % employment change in cell (i, r) , summed across all three themes
$L_{i,r}^0$	Baseline (most recently observed) employment level in cell (i, r)
$\Delta L_{i,r}$	Absolute employment change in cell (i, r) — the dashboard’s primary heatmap output
$\mathcal{K}$	The subset of the 52 sectors with material export exposure (Theme 2’s shock-target set)
d_c^g	Revision to country c ’s IMF growth forecast between consecutive WEO vintages, for the same target year
$w_{k,c}$	Country c ’s share of sector k ’s total Indonesian exports
$g_{i,t}^{yoy}$	Year-on-year growth rate of real retail sales (IPR) for category i at month t
$\bar{g}_{i,t}$	Trailing 36-month moving average of $g_{i,t}^{yoy}$

3 Sector and Region Classification

3.1 The 52-Sector Aggregation

Sectors follow IndoTERM’s own 52-industry aggregation (AGGIND / AGGCOM), itself an aggregation of a 185-sector base classification (IND / COM), with the many-to-one mapping recorded in MIND / MCOM. The source file (`sec52.agg`) is a GEMPACK HAR (Header Array) file — a binary format requiring GEMPACK or the HARpy Python library to parse programmatically; it is *not* a plain-text or CSV file, and generic text/CSV readers will not parse it correctly.

The 52 sectors, in their native order, are listed in Table 2.

Table 2: The 52-sector IndoTERM aggregation.

#	Sector	#	Sector	#	Sector	#	Sector
1	FoodCrops	14	Textiles	27	OtherMan	40	Hotels
2	HortiCrops	15	Leather	28	Electricity	41	Restaurant
3	Estates	16	WoodProd	29	CityGas	42	Telecom
4	Livestock	17	PaperProd	30	WasteMan	43	Finance
5	AgricService	18	NonMetalProd	31	Construction	44	Insurance

Table 2: The 52-sector IndoTERM aggregation.

#	Sector	#	Sector	#	Sector	#	Sector
6	Forestry	19	CoalOilMan	32	VehicTrade	45	OthFinance
7	Fishery	20	Chemical	33	OthTrade	46	FinanSvc
8	Coal	21	Rubber	34	RailTransp	47	RealEstate
9	OilGasGeo	22	BasicMetal	35	LandTransp	48	BusSvc
10	IronOre	23	MetalProd	36	SeaTransp	49	GovAdmin
11	OtherMine	24	Machinery	37	WaterTransp	50	Education
12	FoodMan	25	TranspEquip	38	AirTransp	51	HealthSvc
13	Tobacco	26	Furniture	39	TranspSvc	52	OtherSvc

Codes 1–7 correspond to agriculture/forestry/fishery, 8–11 to mining, 12–27 to manufacturing (16 industries — by far the largest single bucket, and where most of Theme 1’s commodities and most of Theme 3’s mapping decisions concentrate), and 28–52 to utilities, construction, and services (25 industries, almost entirely non-tradable and outside Theme 2’s shock-target set $\mathcal{K}$).

3.2 Provinces

Regions correspond to Indonesia’s 34 provinces, matching the dimension used in BPS’s Inter-Regional Input-Output Table (52 Industries $\times$ 34 Provinces, 2016), which also supplies the baseline employment/output data referenced in Section 9.

4 Theme 1: Commodity Price Shocks

4.1 Mechanism

Eight key commodities, chosen for their combination of (a) large weight in Indonesia’s export or import structure, and (b) geographic concentration making them informative at the province level, are each calibrated as an independent 1% world-price perturbation in IndoTERM.

4.2 Commodity List and Data Sources

Table 3: Theme 1 commodities, reference series, and data access.

Commodity	Reference series	Source	Access
CPO / palm oil	Bursa Malaysia FCPO	Commodities-API; or World Bank Pink Sheet	Paid REST API (real-time) or free monthly XLSX
Coal	Newcastle, kcal/kg	6000 World Bank Pink Sheet	Free monthly XLSX
Nickel	LME cathodes	World Bank Pink Sheet	Free monthly XLSX
Copper	LME grade A	World Bank Pink Sheet	Free monthly XLSX

Table 3: Theme 1 commodities, reference series, and data access.

Commodity	Reference series	Source	Access
TGF (textiles/garments/furniture)	demand-side; no single world price	see Theme 2 for the demand-side construction	—
Electronics	demand-side; no single world price	PHLX Semiconductor Index (SOX); or BLS export price index, FRED series IY3344	Market data API (real-time); or FRED API (free, monthly)
Rubber	Asia TSR 20, SICOM nearby	World Bank Pink Sheet	Free monthly XLSX
Oil/gas	Brent/Dubai/WTI average	World Bank Pink Sheet (monthly); or FRED/EIA (daily)	Free API

The World Bank Pink Sheet (`CMO-Historical-Data-Monthly.xlsx`, fixed URL, updated monthly, no API key) is the single source covering coal, nickel, copper, rubber, and oil/gas — five of the eight commodities — and is the recommended primary source for those five. Oil/gas additionally has a genuine daily-frequency free REST API (FRED or EIA) if higher refresh frequency than monthly is desired for that one commodity specifically.

4.3 Shock Magnitude Construction

On each refresh, the latest available value of the raw driver, $X_{j,t}$, and its value one period prior, $X_{j,t-1}$, are retrieved, and:

$$x_j = 100 \times \frac{X_{j,t} - X_{j,t-1}}{X_{j,t-1}}. \quad (1)$$

Implementation Warning

The formula $x_j = 100 \cdot X_{j,t}/X_{j,t-1}$, without subtracting 1 (equivalently, without subtracting $X_{j,t-1}$ in the numerator), does *not* return zero when $X_{j,t} = X_{j,t-1}$, and will silently corrupt every downstream calculation. **Enforce with a pipeline unit test:** feed a synthetic flat series (no period-over-period change) into the pipeline and assert that $x_j = 0$ exactly. If t and $t-1$ are not calendar years, the chosen frequency must match the frequency used in the elasticity calibration (Section 4.4), and every series should carry an explicit timestamp/frequency tag so mismatched vintages across commodities are visible rather than silently blended.

4.4 Elasticity Calibration

For each of the 8 commodities, IndoTERM is solved once with a 1% perturbation applied to that commodity’s price alone, all other shocks held at zero:

$$\eta_{i,r,j}^{(1)} = e_{i,r,j} \Big|_{x_j=1\%}, \quad j = 1, \dots, 8, \quad (2)$$

producing **8 full** 52×34 **elasticity matrices**. This step is run offline by the user in IndoTERM and is not automated by the data-retrieval pipeline described in this document.

5 Theme 2: Trading-Partner Growth Shocks

5.1 Mechanism

A trading partner’s growth-forecast revision is treated as a uniform, economy-wide demand shifter facing *all* Indonesian exports to that country: a 1 percentage-point (pp) revision to country c ’s growth forecast is assumed to produce a 1% change in demand for any Indonesian export to c (unit pass-through). Product-level differentiation in the resulting shock comes entirely from *which countries buy how much of each product* — i.e., from each sector’s own export destination profile — not from a separately estimated income elasticity per product or per country.

Design Decision

This unit-pass-through assumption is a deliberate simplification, not an estimated parameter. Real income elasticities of import demand are often estimated above 1 (sometimes well above, for industrial/commodity inputs such as nickel or steel). Estimating product/country-specific elasticities from historical Comtrade export values regressed against historical partner GDP is the natural upgrade path (Section 12).

5.2 Step 1: Country Growth Revisions

For each country c covered by the IMF World Economic Outlook (WEO), the forecast revision for the same target year t , comparing the current release to the immediately preceding one, is:

$$d_c^g = g_{c,t}^{(\text{current vintage})} - g_{c,t}^{(\text{prior vintage})}. \quad (3)$$

Data source. IMF World Economic Outlook database, accessed via the IMF’s free SDMX 2.1/3.0 REST API (base URL api.imf.org/external/sdmx/3.0, dataflow WEO, no API key required). Example query construction (Python, via the `sdmx1` library or direct REST calls):

```
import sdmx
IMF_DATA = sdmx.Client('IMF_DATA')
data_msg = IMF_DATA.data('WEO', key='CHN.NGDP_RPCH', params={'startPeriod': '2024'})
```

where NGDP_RPCH is the WEO indicator code for real GDP growth and CHN is the ISO country code (China).

Implementation Warning

WEO is published only **3 times per year** (April, September/October, and a January interim Update). The dashboard’s “press update” refresh **will not see a new value for d_c^g between these three release dates** — this is a genuine constraint of the underlying data, not a limitation of the pipeline. The dashboard UI should indicate the WEO vintage currently in use (e.g., “as of April 2026 WEO”) rather than imply daily freshness for this theme.

5.3 Step 2: Export-Share Weights

For each tradable sector $k \in \mathcal{K}$ (defined in Section 5.5), Indonesia’s export value to its destination countries is retrieved and converted into shares:

$$w_{k,c} = \frac{X_{k,c}}{\sum_{c'} X_{k,c'}}. \quad (4)$$

Data sources (either is viable; both are free and key-based).

- **UN Comtrade:** free API key registration; up to 100,000 records per query and 500 API calls per day on the free tier. Official Python wrapper `comtradeapicall` or R wrapper `comtradr` simplify querying by reporter (Indonesia), partner country, HS code, and flow direction (exports).
- **BPS WebAPI** (`webapi.bps.go.id`): free API key; the `dataexim` resource exposes Indonesia’s own export/import data directly by HS code, in JSON, e.g.

```
https://webapi.bps.go.id/v1/api/dataexim/sumber/{sumber}/kodehs/{kodehs}
/jenishs/{jenishs}/tahun/{tahun}/periode/{periode}/key/{key}
```

with an official Python wrapper (`stadata`) available.

Design Decision

Export-share weights $w_{k,c}$ change slowly (annual trade patterns, not month-to-month), so this step does **not** need to be re-run on every dashboard refresh. Recompute annually (or whenever a new year of Comtrade/BPS trade data becomes available) and cache; only d_c^g (Step 1) needs to be checked against the live WEO release schedule on every refresh.

5.4 Step 3: Aggregation into a Sector-Specific Shock

The shock magnitude for sector k is the export-share-weighted average of the growth-forecast revisions facing its actual buyers:

$$x_k = q_k^T = \sum_c w_{k,c} \cdot d_c^g. \quad (5)$$

Derivation note. An earlier, more general formulation considered a separate income elasticity $\epsilon_{i,c}$ per (product, country) pair, giving $q_{i,c} = \epsilon_i \cdot d_c^g$ aggregated as $q_i^T = \sum_c w_{i,c} \epsilon_i d_c^g = \epsilon_i \sum_c w_{i,c} d_c^g$. Under the unit-pass-through assumption ($\epsilon_i \equiv 1$ for all products, applied uniformly across countries rather than varying by country), this collapses exactly to Equation (5): there is no need to ever materialize a per-country shock $q_{k,c}$ individually, only the weighted average.

5.5 The Tradable Sector Set $\mathcal{K}$

Not all 52 sectors have meaningful export exposure; shocking sectors with near-zero export shares would add CGE-calibration cost for negligible benefit. $\mathcal{K}$ is determined empirically from the export column of the 52-sector Input-Output table (export value $\div$ total output per sector, against a chosen threshold — 1% or 5% are reasonable candidates; the exact cutoff is a user decision not yet finalized).

Implementation Warning

The sector list below is a **category-level estimate** based on which of the 52 sectors are agriculture/mining/manufacturing (vs. services, utilities, construction, which are excluded with high confidence). It has not yet been verified row-by-row against actual export/output ratios in the IO table. Treat as a starting point for verification, not a final list.

Candidate $\mathcal{K}$ (by sector number and name, from Table 2): 3 (Estates), 7 (Fishery), 8 (Coal), 9 (OilGasGeo), 10 (IronOre), 11 (OtherMine), 12 (FoodMan), 14 (Textiles), 15 (Leather), 16 (WoodProd), 19 (CoalOilMan), 20 (Chemical), 21 (Rubber), 22 (BasicMetal), 23 (MetalProd), 24 (Machinery), 25 (TranspEquip), 26 (Furniture) — approximately **17–18 sectors**.

5.6 Elasticity Calibration

For each sector $k \in \mathcal{K}$, IndoTERM is solved once with a 1% export-demand perturbation to sector k alone:

$$\eta_{i,r,k}^{(2)} = e_{i,r,k} \Big|_{x_k=1\%}, \quad k \in \mathcal{K}, \quad (6)$$

producing **approximately 17 full** 52×34 **elasticity matrices**.

6 Theme 3: Domestic Demand Shocks

6.1 Mechanism and Motivation

Themes 1 and 2 are both externally focused (world prices, foreign growth) and so are silent on purely domestic demand conditions — relevant for sectors selling mainly to Indonesian consumers (retail, domestically-oriented manufacturing, food processing for the domestic market). Theme 3 fills this gap using Bank Indonesia’s real (volume) retail sales index.

6.2 Why a Real (Volume) Index, Not a Price Index

An earlier design considered Indonesia’s Consumer Price Index (CPI) as the domestic-demand signal. This was rejected: a price index alone cannot distinguish a demand-driven price change from a supply/cost-push price change — a price rise is consistent with either rising demand (price up, quantity up) or a negative supply shock (price up, quantity down), and these have opposite implications for layoff risk. The originally proposed resolution was a price–quantity co-movement check (pairing CPI with a volume series and only treating a price move as demand-driven when price and quantity move in the same direction). This check was **dropped** once the design moved to Bank Indonesia’s *Indeks Penjualan Riil* (IPR, “Real Sales Index”): IPR is already a deflated, real/volume measure, so the price/demand identification problem that motivated the co-movement check does not arise from this series in the first place, and the extra CPI-fetch step would be solving an already-solved problem.

Implementation Warning

Accepted, transparent simplification: this reasoning assumes Bank Indonesia’s internal deflation of IPR is adequate. If BI’s deflator imperfectly strips out category-specific price effects (e.g. uses a general rather than category-specific price index internally), some residual price-driven variation could still leak into the yoy growth figure used as Theme 3’s shock.

This is not independently re-verified against CPI in this version of the design — a contestable but openly stated choice, not an oversight.

6.3 Data Source

Bank Indonesia’s Survei Penjualan Eceran (SPE, Retail Sales Survey), specifically its headline output, the Indeks Penjualan Riil (IPR). Published monthly. There is no key-based REST API; data is distributed as a downloadable zip file attached to each month’s release page, at a predictable URL pattern:

```
https://www.bi.go.id/id/publikasi/laporan/Documents/  
Data-Series-SPE-{Month}-{Year}.zip
```

where {Month} is the Indonesian month name (e.g. **Februari**) and {Year} the four-digit year. The release page itself follows the analogous pattern `.../Pages/SPE-{Month}-{Year}.aspx`.

6.4 “Press Update” Logic for Theme 3

1. Construct the expected URL for the current month/year using the pattern above.
2. Issue a HEAD request (or attempt a fetch with graceful fallback) to check whether this URL exists and differs from the last cached vintage.
3. If a newer month’s zip is available: download, unzip, extract the category-level yoy growth series, append to the cached historical panel.
4. If not (i.e. the current month’s release has not yet been published): proceed using the most recently cached vintage; do not error.

6.5 IPR Categories

IPR publishes seven categories, retained in full in the data pipeline regardless of whether each one is ultimately wired into an elasticity (see Section 6.9):

1. Makanan, Minuman, dan Tembakau (Food, Beverages, Tobacco)
2. Sandang (Clothing/Textiles)
3. Suku Cadang dan Aksesori (Automotive Spare Parts and Accessories)
4. Bahan Bakar Kendaraan Bermotor (Motor Vehicle Fuel)
5. Barang Budaya dan Rekreasi (Cultural and Recreational Goods)
6. Peralatan Informasi dan Komunikasi (Information and Communication Equipment)
7. Perlengkapan Rumah Tangga Lainnya (Other Household Equipment)

6.6 Why Year-on-Year, Not Month-to-Month

IPR publishes growth both year-on-year (yoy) and month-to-month (mtm). **Only the yoy figure is used.** Indonesian retail sales carry strong, recurring seasonality tied to the national religious/holiday calendar (Ramadan, Idulfitri, Christmas/New Year), and these dates shift by approximately 11 days earlier each Gregorian year for the lunar-calendar holidays. Comparing month t to month $t - 12$ (yoy) automatically cancels this seasonality; compounding mtm growth into an annualized rate does not, and will badly overstate or understate the true signal whenever a holiday period falls in the observation window. (BPS/BI press releases routinely and explicitly attribute large mtm swings to specific holiday periods, confirming the size of this seasonal effect.)

6.7 Benchmark/Trend and Shock Construction

Raw yoy growth is always technically signed correctly (it can be negative), but a positive-yet-low growth rate still represents a meaningful *slowdown* relative to a sector’s normal trend, and a layoff-risk signal needs to register that as a negative shock — not as “still growing, so still fine.” The shock is therefore constructed as a **deviation from a trailing benchmark**, not the raw yoy growth rate itself:

$$\bar{g}_{i,t} = \frac{1}{36} \sum_{k=1}^{36} g_{i,t-k}^{yoy}, \quad x_i = g_{i,t}^{yoy} - \bar{g}_{i,t}. \quad (7)$$

Design Decision

A trailing 36-month moving average was chosen over (a) a fixed historical base-period mean, which requires an arbitrary choice of reference window and does not adapt to genuine structural shifts (e.g. permanent changes in e-commerce penetration affecting the ICT-equipment category), and (b) formal trend extraction (HP filter or similar), judged disproportionate for a first-version dashboard and subject to end-point bias exactly where the signal is most needed. The 36-month window is long enough to average out holiday-timing noise across different years while remaining short enough to track genuine multi-year drift.

Implementation Warning

$\bar{g}_{i,t}$ requires **at least 36 months of historical data before it can be computed for the first time**. The initial pipeline setup must backfill several years of past SPE zip files (not just the current month) before Theme 3 can produce any output. Each subsequent monthly refresh only needs to add one new observation and slide the window forward.

6.8 Regional Dimension: National Shock, Regional Elasticity

IPR’s underlying survey covers a limited set of cities, not all 34 provinces, so x_i is constructed and used as a **single national value per category** — there is no $x_{i,r}$ varying by province. All regional differentiation in Theme 3’s contribution to the heatmap comes entirely from the elasticity tensor $\eta_{i,r,m}^{(3)}$, which *is* fully calibrated at 52×34 resolution. This means a national slowdown in, say, domestic clothing sales is assumed to affect every province’s clothing-sector employment in the same relative direction, with only the magnitude of that effect varying by province (through η) — genuine sub-national divergence in domestic demand itself (one province softening while another strengthens) is not captured by this version.

Design Decision

This national-shock/regional-elasticity structure is not unique to Theme 3 — Theme 1 (a world commodity price) and Theme 2 (a country’s growth forecast) are also necessarily national-level shocks, since neither has an Indonesian regional dimension to begin with. All three themes therefore share the same underlying structural pattern: a single national shock magnitude, multiplied against a fully region-differentiated elasticity tensor.

6.9 Mapping IPR Categories to the 52-Sector Classification

This mapping is performed manually by the model’s developer, not automated by the data pipeline, since several IPR categories are diffuse enough to require judgment (and some may legitimately split across more than one of the 52 sectors). Indicative mappings, based on category content alone:

- Makanan, Minuman, dan Tembakau → FoodMan (12), possibly Tobacco (13)
- Sandang → Textiles (14), possibly Leather (15)
- Suku Cadang dan Aksesoris → TranspEquip (25)
- Bahan Bakar Kendaraan Bermotor → OilGasGeo (9) and/or CoalOilMan (19)
- Barang Budaya dan Rekreasi → diffuse; candidate sectors include OtherMan (27), OthTrade (33), or a services sector, pending the developer’s judgment
- Peralatan Informasi dan Komunikasi → Machinery (24) (where electronics is aggregated in this classification)
- Perlengkapan Rumah Tangga Lainnya → Furniture (26), possibly splitting with other manufacturing sectors

6.10 Elasticity Calibration: All 52 Sectors

Design Decision

Unlike Theme 2, where the elasticity-target set was restricted to $\mathcal{K}$ (tradable sectors only) because non-tradable sectors have negligible export-demand exposure by construction, Theme 3’s elasticity target set is **all 52 sectors, with no pre-filtering by IPR’s current category coverage**. The reasoning: every sector sells *something* domestically, even sectors IPR’s seven categories do not currently observe; whether a sector’s domestic-demand elasticity turns out to be economically negligible is itself a CGE result to be discovered, not an assumption to be imposed in advance based on today’s data availability. Calibrating all 52 up front means that if more granular domestic-demand data becomes available later (e.g. from Susenas or an expanded retail survey), the elasticity side of the pipeline is already complete and does not block using it.

For each of the 52 sectors, IndoTERM is solved once with a 1% domestic-demand perturbation to that sector alone:

$$\eta_{i,r,m}^{(3)} = e_{i,r,m} \Big|_{x_m=1\%}, \quad m = 1, \dots, 52, \quad (8)$$

producing **52 full** 52×34 **elasticity matrices** — the largest single calibration task among the three themes.

7 Why Elasticities Are Theme-Specific

A central design decision, worth stating explicitly because it substantially affects the total CGE-calibration workload: **a sector’s elasticity tensor is not shared across themes**, even when the same sector is the directly-shocked target under more than one theme (e.g. Textiles appears in Theme 1 as an export-price target and, via the Sandang mapping, in Theme 3 as a domestic-demand target).

The justification is economic, not merely a bookkeeping convention: a domestic-demand shock and an export-price shock to the same sector are mechanically different perturbations to In-DoTERM. They enter the model through different channels (domestic final consumption vs. export price/demand), and they can plausibly activate different substitution elasticities and different general-equilibrium feedback paths as they propagate through the input-output structure — domestic consumers and export markets need not substitute toward or away from a sector’s output in the same way. Treating $\eta_{i,r,\text{Textiles}}^{(1)}$ and $\eta_{i,r,\text{Textiles}}^{(3)}$ as identical would be assuming away a real distinction rather than testing or justifying it.

The consequence is a larger total calibration workload than a shared-elasticity design would require, summarized in Table 4.

Theme	# of shocks	# of 52×34 matrices	Calibration basis
1 — Commodity price	8	8	1% price perturbation, 8 commodities
2 — Partner growth	≈ 17 ($\mathcal{K}$)	≈ 17	1% export-demand perturbation, tradable sectors
3 — Domestic demand	52	52	1% domestic-demand perturbation, all sectors
Total		≈ 77	

Table 4: Elasticity-calibration scope across the three themes. This is the CGE-modeling workload, supplied externally by the model developer; it is not automated by the data-retrieval pipeline described in this document.

8 Aggregation Across Themes

Each theme’s contribution to the percentage employment change in cell (i, r) is:

$$E_{i,r}^{(1)} = \sum_{j=1}^8 \eta_{i,r,j}^{(1)} \cdot x_j, \quad (9)$$

$$E_{i,r}^{(2)} = \sum_{k \in \mathcal{K}} \eta_{i,r,k}^{(2)} \cdot q_k^T, \quad q_k^T = \sum_c w_{k,c} \cdot d_c^g, \quad (10)$$

$$E_{i,r}^{(3)} = \sum_{m=1}^{52} \eta_{i,r,m}^{(3)} \cdot (g_{m,t}^{yoy} - \bar{g}_{m,t}). \quad (11)$$

The total percentage employment effect is the simple sum across themes:

$$E_{i,r} = E_{i,r}^{(1)} + E_{i,r}^{(2)} + E_{i,r}^{(3)}. \quad (12)$$

This inherits the additive-separability assumption at two nested levels: across shocks *within* a theme (Equations (9)–(11)), and across the three *themes* themselves (Equation (12)). Both levels assume no interaction effects between simultaneously active shocks — discussed further in Section 11.

9 Severity-Weighted Heatmap

$E_{i,r}$ alone is a *percentage* change, and so weights a large percentage shock to a small sector–province cell identically to the same percentage shock hitting a large cell — the wrong lens for a *risk* dashboard, which should foreground absolute headcount exposure. The primary heatmap output is therefore the absolute employment change:

$$\Delta L_{i,r} = E_{i,r} \cdot L_{i,r}^0, \quad (13)$$

where $L_{i,r}^0$ is the baseline (most recently observed) employment level in cell (i,r) , sourced from the same Input-Output table that supplies the sector and province dimensions (Section 3), or from a more frequently updated labor-force survey if available — in which case the vintage difference between $L_{i,r}^0$ ’s source and the SAM’s base year should be tagged explicitly in metadata, since using two different-vintage sources silently is a hidden inconsistency.

$\Delta L_{i,r}$ is rendered as the dashboard’s primary heatmap (color intensity proportional to magnitude, with sign indicating job gains vs. losses); $E_{i,r}$ (the pure percentage figure) is retained as a secondary/tooltip layer, since “where is the biggest percentage shock” and “where is the biggest headcount exposure” are different and both useful questions for a policy user to be able to ask.

10 Computational Architecture: “Press Update” Logic

1. **Theme 1.** For each of the 8 commodities, check the latest available value from its source (World Bank Pink Sheet monthly file, or FRED/EIA daily API for oil/gas) against the last cached value. If newer, fetch and recompute x_j via Equation (1).
2. **Theme 2.** Check whether a new IMF WEO vintage has been published since the last cached release (April / September-October / January). If so, re-pull d_c^g for all relevant countries via the IMF SDMX API and recompute q_k^T for each $k \in \mathcal{K}$ using Equation (5), combined with the cached export-share weights $w_{k,c}$ (these are recomputed on a much slower, e.g. annual, cadence from Comtrade/BPS trade data — not on every refresh).
3. **Theme 3.** Construct the expected SPE zip URL for the current month; check existence/freshness against the cached vintage; if newer, download, unzip, append the new month’s category-level yoy figures to the trailing panel, and recompute $\bar{g}_{i,t}$ and x_i for all seven IPR categories via Equation (7).
4. **Multiply and aggregate.** Each theme’s shock vector is multiplied against its respective pre-stored elasticity tensor (a tensor contraction, not a CGE solve), summed across themes (Equation (12)), and scaled by baseline employment (Equation (13)).
5. **Render.** The resulting $\Delta L_{i,r}$ matrix is rendered as the heatmap; $E_{i,r}$ is exposed as supplementary detail.

Design Decision

The elasticity tensors themselves (Section 7, the slow and expensive part of the whole pipeline) are **never recomputed on refresh**. They are supplied externally by the model developer from IndoTERM runs and only need to be re-supplied when the CGE model, its closures, or the base-year SAM change — not on any dashboard-refresh cadence.

11 Assumptions and Known Limitations

- **Local linearity.** Each $\eta_{i,r,j}^{(\theta)}$ is the slope measured at a 1% perturbation, applied as though constant for any observed magnitude of x_j . CGE responses are generally nonlinear once closures bind differently at different shock sizes — for instance, the dual labor-market closure (rigid-wage/quantity-adjusting formal manufacturing vs. underemployment-absorbing informal sectors) may not respond proportionally as shock size grows. Historical shocks of genuine interest (large coal, CPO, or nickel price swings; sizable WEO forecast revisions during global downturns) are frequently far outside the small neighborhood in which a 1%-calibrated slope is guaranteed to extrapolate cleanly.
- **Additive separability, within and across themes.** Summing across shocks within a theme, and summing across the three themes (Section 8), both assume the absence of interaction effects. This holds only approximately under genuine joint general-equilibrium market clearing, and the approximation degrades when multiple shocks load on the same factor or sector simultaneously — e.g. Theme 1’s TGF export-price shock and Theme 3’s Sandang domestic-demand shock both affecting Textiles employment in the same period, with no cross-term capturing their joint effect.
- **Theme-specific elasticities, no cross-theme reuse** (see Section 7 for full justification). This roughly triples the CGE-calibration workload relative to a design that shares one elasticity tensor per sector across all themes, but avoids silently assuming that domestic-demand and export shocks propagate identically.
- **Theme 3’s shock is national, not regional** (Section 6.8). Genuine sub-national divergence in domestic demand — one province’s retail sector softening while another’s strengthens — is not captured; only the elasticity side of Theme 3 varies by province.
- **IPR’s internal deflation is taken at face value.** If Bank Indonesia’s real-sales deflator imperfectly isolates volume from price effects, some residual price-driven noise could leak into Theme 3’s shock construction. This is an openly stated simplification, not independently re-verified against CPI in this version.
- **Unit pass-through in Theme 2.** A 1pp growth-forecast revision is assumed to produce exactly a 1% export-demand change, uniformly across products and countries, with no separately estimated income elasticity. True elasticities frequently differ from 1, especially for industrial/commodity exports.
- **Time-invariant elasticities and baseline employment between refreshes.** Both $\eta_{i,r,j}^{(\theta)}$ and $L_{i,r}^0$ are held fixed until the user manually recalibrates IndoTERM or re-sources employment data; an explicit recalibration cadence for each, and explicit vintage tagging where sources differ, should be defined rather than left implicit.
- **Heterogeneous epistemic status of inputs across themes.** Theme 1’s inputs are CGE-simulated structural elasticities applied to directly observed world prices. Theme 2 combines a CGE-simulated elasticity with an unestimated unit-pass-through assumption. Theme 3 combines a CGE-simulated elasticity with an empirical trend-deviation measure subject to a deflation-quality caveat. A dashboard user should not be given the impression that all three carry the same epistemic weight.

- **Discrete, low-frequency refresh for Theme 2 specifically.** Unlike Themes 1 and 3, which can in principle update monthly or even daily, Theme 2’s d_c^g only changes 3 times per year (WEO release schedule) — the dashboard should communicate this vintage explicitly rather than implying continuous freshness.
- **No joint re-solve benchmarking yet.** There is currently no periodic check of the full linear-additive approximation (Equation (12)) against a true joint CGE re-solve under realistic, simultaneously active multi-theme shock combinations. Before any high-stakes policy communication, this benchmark should be run to establish an empirical error bound on the approximation.

12 Future Work

- **Retire the linear shortcut entirely, rather than patch it.** The calibrate-once-and-multiply design (Sections 4–10) is a deliberate stopgap, not the intended end state. Once IndoTERM can be run live on a server with acceptable latency, the dashboard should move from *look up a pre-calibrated elasticity and multiply to solve the full CGE on the actual observed combination of shocks on every refresh*. This removes the linearity and additive-separability assumptions in Section 11 outright, rather than approximating around them with interaction terms or quadratic corrections. Server sizing, solve time, and a queuing/caching strategy for concurrent users are the open questions for that transition, to be scoped as a separate piece of work.
- **Two-point calibration check for Theme 1.** For commodities with historically large price swings (coal, CPO, nickel), additionally calibrate at a 10% perturbation and compare the implied slope to the 1%-calibrated η . If stable, the linear extrapolation is empirically supported for that commodity; if not, fit a quadratic correction term using the two calibration points.
- **Estimate product/country-specific elasticities for Theme 2** in place of the unit-pass-through assumption, via a log-log regression of historical Comtrade export values against historical partner GDP growth, by product.
- **Sub-national domestic-demand data for Theme 3**, if/when a province-level consumption proxy becomes available (e.g. BPS’s Susenas household expenditure survey) — though Susenas’s much lower publication frequency relative to IPR’s monthly cadence would require its own treatment, not a simple substitution.
- **Interaction terms across themes**, for sectors confirmed to be jointly and materially hit by more than one theme simultaneously, once the re-solve benchmarking in Section 11 quantifies how large the currently-omitted interaction error actually is in practice.

13 Open Items Requiring Further User Input

- Verify $\mathcal{K}$ (Section 5.5) row-by-row against actual export/output ratios in the 52-sector Input-Output table; the list given is a category-level estimate, not yet confirmed.
- Choose the export-share threshold defining $\mathcal{K}$ (1%? 5%? another value?) — not yet decided.

- Complete the manual mapping of IPR's seven categories onto the 52-sector classification (Section 6.9); some categories may legitimately split across more than one sector.
- Specify the source and vintage for $L_{i,r}^0$ (baseline employment); not yet finalized, and should be tagged with its own update cadence, potentially distinct from the IndoTERM SAM's base year.